

Q1

We know that $\frac{\partial f_1}{\partial x_1}(x) = -2(A_{11}x_1 + A_{12}x_2)$ and that $\frac{\partial f_2}{\partial x_2}(x) = -2(B_{21}x_1 + B_{22}x_2)$

So,

$$x_1^{k+1} = x_1^k - 2\alpha(A_{11}x_1^k + A_{12}x_2^k)$$

$$x_2^{k+1} = x_2^k - 2\alpha(B_{21}x_1^k + B_{22}x_2^k)$$

We can rewrite this as

$$x^{k+1} = \begin{pmatrix} x_1^{k+1} \\ x_2^{k+1} \end{pmatrix} = \begin{pmatrix} 1 - 2\alpha A_{11} & -2\alpha A_{12} \\ -2\alpha B_{21} & 1 - 2\alpha B_{22} \end{pmatrix} \begin{pmatrix} x_1^k \\ x_2^k \end{pmatrix} = (I - 2\alpha C) \begin{pmatrix} x_1^k \\ x_2^k \end{pmatrix}$$

where $C = \begin{pmatrix} A_{11} & A_{12} \\ B_{21} & B_{22} \end{pmatrix}$

We know that it will diverge if the spectral radius is above 1 for all positive α for $D(\alpha) = I - 2\alpha C$.

Since the eigenvalues of $D(\alpha)$ are $\mu_i = 1 - 2\alpha\lambda_i$, where λ_i 's are the eigenvalues of matrix C .

Because α is positive, if one λ_i is negative, then $|\mu_i| > 1$

An easy way to do this is to see that if we can find two positive definite matrixes A and B so that the matrix C based on the first and second row of A and B can have negative eigenvalues.

So, if $A = \begin{pmatrix} 1 & 3 \\ 3 & 10 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$, then A is positive definite since $A_{11} = 1 > 0$, and

$\det A = 10 - 9 = 1 > 0$; B is also positive definite since $B_{11} = 1 > 0$, and

$\det B = 5 - 4 = 1 > 0$.

Plugging A and B to formulate C, we have $C = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$, whose 2 eigenvalues λ_i 's are 6.1623 and

-0.1623, the negative λ_- making one $\mu_- = 1 + 2 * 0.1623 = 1.3246 > 1$, then we know that the iteration matrix has a spectral radius (smallest eigenvalue) larger than 1, this must diverge.

Q2

For the Even player, if we know that y is fixed and at Nash Equilibrium, then maximizing $x^T Ay$ subject to the basic weight/probability restraints is equivalent to

$$\max_x \sum_i x_i (Ay^*)_i$$

Because we know this is a mixed strategy, if there were one coordinate (for example $(Ay^*)_k$ for some index k), then the player Even would place all of its chips (x_k) into that one ($x_k = 1$), which would no longer make it a mixed strategy, but a pure strategy.

Therefore, for a mixed strategy (multiple $x_i > 0$), we know that

$$(Ay^*)_1 = \dots = (Ay^*)_n = \alpha$$

So by vector operations, $Ay^* = \alpha e$

For the Odd player, if we know that x is fixed and at Nash Equilibrium, then maximizing $-x^T Ay$ subject to the basic weight/probability restraints is equivalent to

$$\max_y \sum_i y_i (-A^T x^*)_i$$

Because we know this is a mixed strategy, if there were one coordinate (for example $(-A^T x^*)_k$ for some index k), then the player Odd would place all of its chips (y_k) into that one ($y_k = 1$), which would no longer make it a mixed strategy, but a pure strategy.

Therefore, for a mixed strategy (multiple $x_i > 0$), we know that

$$(-A^T x^*)_1 = \dots = (-A^T x^*)_n = -\beta$$

So by vector operations and cancelling the negative signs on both sides, $A^T x^* = \beta e$

By expanding the matrix-vector product, this matrix vector expression is better simplified as the proof of 1) $Ax^* = \bar{f}e$, and 2) $e^T x^* = 1$, in vector form.

Since we know that the Nash Equilibrium is when $F(x^*) = 0$, or when

$$\forall i, 0 = x_i^* ((Ax^*)_i - (x^*)^T Ax^*) \rightarrow (Ax^*)_i = (x^*)^T Ax^*$$

And since we know by definition that

$$\bar{f} = \bar{f}(x^*) = (x^*)^T Ax^*$$

So, $(Ax^*)_i = \bar{f}$, and since this applies for all i , we can vectorize it into:

$$Ax^* = \bar{f}e \tag{1}$$

As the replicator dynamics are defined on the simplex with constraint $e^T x = 1$, the stationary point x^* by definition of the state space must satisfies:

$$e^T x^* = 1. \tag{2}$$

Combining the above 2 proved equations, we thus proved the matrix equation.

Now prove there is at most one equilibrium by contradiction: Assume there are at least 2 different interior equilibria x^*, y^* with corresponding $\bar{f}_x$ and $\bar{f}_y$ (generic function with 614 methods), then we have

$$Ax^* = \bar{f}_x e, e^T x^* = 1,$$

$$Ay^* = \bar{f}_y e, e^T y^* = 1,$$

Taking the differences of the above equations:

$$Az = b e, e^T z = 0, \tag{3}$$

where $z = x^* - y^*$ which is not 0 by assumption, and $b = \bar{f}_x - \bar{f}_y$.

Now consider the cases for b :

1. if $b \neq 0$, then by (3) $e = Az/b$, which belongs to $\text{Range}(A)$, contradicting the assumption $e \notin \text{Range}(A)$
2. if $b = 0$, the (3) turns into:

$$Az = 0, e^T z = 0 \quad (4)$$

by assumption, $z \neq 0$ and $z \in \ker A$. As we knew $\dim(\ker A) \leq 1$, any nonzero vector in $\ker A$ spans the null space.

we know that

$$\text{Range}(A^T) = (\ker A)^\perp$$

and given that $e \notin \text{Range}(A^T)$, so e is not orthogonal to $\ker A$, thus, for any nonzero $a \in \ker A$, $e^T a \neq 0$, but we knew $z \in \ker A$ and z is nonzero, we must have $e^T z \neq 0$, contradicting $e^T z = 0$ in (4). so case 2 also causes contradiction.

We thus proved that it is impossible to have two or more equilibria, therefore we can have at most one equilibrium, which concludes the proof of this question.